

Sean Lee

Translational Oncology & Computational Biology Researcher

Katy, Texas | [linkedin.com/in/sean-lee-ab5645383](https://www.linkedin.com/in/sean-lee-ab5645383)

PROFESSIONAL PROFILE

Rising junior at Cinco Ranch High School focused on translational oncology and computational biology. Began independent cancer research at age 14; has served as first author on peer-reviewed work and collaborates with an Associate Professor at MD Anderson Cancer Center and researchers at Baylor College of Medicine.

Research spans biliary tract cancer, hepatocellular carcinoma, NSCLC radiomics, multi-omic tumor-microenvironment analysis, statistical modeling, and survival modeling.

Current work maps how pSTAT3—an inflammatory-cytokine-driven signaling pathway—reshapes the tumor microenvironment across esophageal, gastric, colorectal, pancreatic, liver, and cholangiocarcinoma. The project stratifies TCGA patients by pSTAT3 protein level and integrates genomic, transcriptomic, and spatial microenvironment data to test whether the axis determines immune landscape, fibrotic architecture, and patient survival across gastrointestinal malignancies.

Founded a 120+ member high-school cancer research organization, co-founded a nonprofit addressing cancer-related financial toxicity in collaboration with the MD Anderson Financial Clearance Center, and runs a photography business serving 25+ clients. Motivated by closing the gap between data and clinical impact.

RESEARCH & ENTREPRENEURIAL EXPERIENCE

Co-Founder & CTO | *Sutura Genomics, Inc.*

Apr 2026–Present

Houston metro area, Texas

- Co-founded an AI platform that reconstructs torn, folded, deformed, or partially missing tissue sections for spatial-biology workflows; applied to Y Combinator's Fall 2026 batch.
- Company and live demo: sutura-genomics.bio. Product description used in the YC application: "AI that fixes torn tissue data for spatial biology."
- Co-founders have known each other for seven years after meeting at a fifth-grade tennis tournament; progressed from shared science experiments in seventh grade to USABO, International Biology Bowl, Science Olympiad, and computational-biology research collaboration.
- Built the cross-donor evaluation pipeline and ground-truth methodology, drawing on 2.5 years of computational oncology research at MD Anderson; both founders write all production code, with no non-founder technical contributors.
- Reached four active research labs using Sutura on live tissue datasets; professors, PhD students, and research staff evaluate alignment quality, workflow automation, and usability.
- Generated \$13,500 in June 2026 revenue entirely from software: a \$12,500 two-month research-lab subscription and a \$1,000 paid pilot for an independent PhD researcher covering three datasets; \$0 consulting revenue.
- Advanced late-stage workflow-integration projects with AdvaitaBio and BioChain on internal or customer datasets and entered early commercial discussions with two additional spatial-biology companies.
- Built a 35-researcher waitlist from 60+ pre-build interviews across academia, biotechnology, and pharmaceuticals; onboarding labs gradually while validating cross-donor generalization on external datasets.
- Shipped a live web product and interactive demo, a bioRxiv methodology preprint, four lab deployments, the first two paying customers, and commercial integration work within eight days of going full-time.
- Developed a hybrid alignment system that automatically selects the best method for each dataset and consistently outperforms individual approaches in internal benchmarks; Sutura reduced pixel-level alignment error by approximately an order of magnitude across synthetic and real-world tear scenarios.
- Positioning Sutura as the vendor- and tissue-agnostic reconstruction layer between spatial-imaging hardware and downstream analysis software, replacing hours of manual correction or costly sample repetition when legacy methods such as PASTE2, STalign, and GPSA fail.
- Works with scientific advisor Dr. Sunyoung S. Lee, MD Anderson GI Medical Oncology faculty, who provides clinical guidance and manuscript review but does not write code.
- Began full-time work June 25, 2026, averaging approximately 12 hours per day; incorporated June 27, 2026 as a Delaware C-Corporation through Stripe Atlas. Current company location is Texas, with plans to relocate to San Francisco after YC.
- Technical stack: Python 3.11, PyTorch, PyTorch Geometric, Scanpy, AnnData, scikit-learn, SciPy, Next.js, TypeScript, Three.js, Netlify, Git/GitHub, Claude Code, and Claude models.
- Business model: paid pilots converting to annual subscriptions based on data volume; customer interviews indicate academic-lab pricing of \$5,000–\$10,000 annually and biotechnology/pharma deployments of \$50,000–\$200,000+ annually.
- Equity structure reported in the YC application: Sean Lee, CTO and co-founder, 45%; Rushil Maniar, CEO and co-founder, 45%; employee/advisor option pool, 10%; no outside investment and not fundraising at the time

of application.

Research Intern | *UT MD Anderson Cancer Center*

Jan 2024–Present

Remote

- Conduct computational cancer-genomics research under a GI medical oncologist and associate professor at MD Anderson—the core of the researcher's work.
- Uses an open MD Anderson clinical and molecular dataset that is not publicly available, providing a rare resource for high-school-level research.
- Writes code, builds models, and performs patient-genomic-data analysis, including pSTAT3 signaling and the tumor microenvironment across six gastrointestinal cancer cohorts, with discovery and validation on public cohorts including TCGA and CPTAC.

Clinical Research Apprentice | *UT MD Anderson Cancer Center*

Sep 2024–Oct 2024

- Completed a one-on-one apprenticeship shadowing a top 1% GI medical oncologist at MD Anderson and observing the translational side of the researcher's computational work.
- Traveled with the faculty mentor to the DAVANCO conference in Bermuda, an invitation-only meeting where one MD Anderson faculty member is invited each year.

President & Founder | *Onco Research Scholars*

Jan 2026–Present

- Mentors students through the research application process, from developing research ideas to cold-emailing founders, and leads a student organization.
- Built cancer-research opportunities through full-process mentoring, professor/PI cold-emailing, and relationship development; 25+ members secured research positions at Stanford, Yale, MIT, MD Anderson, Google, and UT across psychology, neuroscience, cancer, and virology, from incoming freshmen to college undergraduates.
- Founder of an international research incubator, coordinating student oncology cohorts across 3 continents and 10 states

Researcher | *Cholangiocarcinoma Foundation*

Jan 2026–Present

Salt Lake City, Utah · Remote · Seasonal

- Conducts statistical modeling and survival modeling.

Researcher | *European Society for Medical Oncology (ESMO)*

Apr 2026–Present

Singapore · Remote · Seasonal

Co-Founder & CFO | *ThroughWay*

Apr 2026–Present

Katy, Texas · Hybrid

- Co-founded a nonprofit reducing financial toxicity and improving insurance access for cancer patients; mentored by the MD Anderson Financial Clearance Center.
- Uses MD Anderson and Cholangiocarcinoma Foundation connections to expand partnerships and the donor network, with growth planned around the Cholangiocarcinoma Foundation 2027 Annual Conference.

PUBLICATIONS & RESEARCH OUTPUT

Tissue tearing degrades optimal-transport registration of spatial transcriptomics beyond displacement magnitude: a multiseed deformation benchmark and a supervised graph cross-attention proof-of-concept

[bioRxiv](#) Jul 5, 2026

Nab-paclitaxel Across Four Registrational Phase 3 Trials in Advanced Gastrointestinal Cancers: A Complete-Enumeration Meta-Analysis Bounding the Survival Effect, and a Tested Non-Replication of the pSTAT3 Biomarker Signal

[OSF Registries](#) — [manuscript under review at CCC](#) Jul 1, 2026

Solo-authored systematic review and meta-analysis of the complete registrational Phase 3 nab-paclitaxel program (4 trials; 3,383 patients) in advanced GI cancers; preregistered on OSF.

A 6-Gene pSTAT3 Transcriptomic Score Identifies an Immunosuppressive, Chemotherapy-Resistant Phenotype and Predicts Poor Survival in Biliary Tract Cancer

[The Oncologist](#) — [Oxford University Press](#) Jun 23, 2026

Visible author line begins: Sean GP Lee, Mohamed Nuh, Monica Hsiang, Akhila Madulapalli Reddy, Sunyoung...

A pan-cancer identifiability atlas for compartment-resolved pathway attribution from bulk transcriptomes

[Open Science Framework \(OSF\)](#)

Competing risks are ignored in most SEER cancer-specific-mortality nomograms: a systematic audit with real-patient reanalysis and a validated correction pipeline

[Open Science Framework \(OSF\)](#)

Limits of a satellite bloom-timing surrogate for match–mismatch: a global recruitment test and a regional prey-timing validation

[Open Science Framework \(OSF\)](#)

Most published immunotherapy-response gene signatures do not outperform random gene sets, and collapse onto a single tumor-inflammation axis

[Open Science Framework \(OSF\)](#)

Tumor Phospho-STAT3 Marks a Shared Angiogenic-Fibrotic, Immune-Replete Microenvironment Across Gastrointestinal Cancers: A Multi-Omic Analysis of 1,274 Tumors

[First-author manuscript](#) — [under review](#)

When can a pathway program be attributed to a tumour compartment from bulk RNA? An identifiability audit across immune-checkpoint cohorts

[Open Science Framework \(OSF\)](#)

HONORS, PRESENTATIONS & CONFERENCES

Youngest presenter—and only high schooler/undergraduate—in the documented history of the Cholangiocarcinoma Foundation Annual Conference | *Cholangiocarcinoma Foundation Annual Conference 2026* Issued Apr 2026

Associated with UT MD Anderson

- Received travel support through an MD Anderson grant and presented first-author pSTAT3 transcriptomic-scoring research to an international panel of gastrointestinal oncology experts as the youngest participant in the foundation's history.

Poster Presenter, ESMO Asia Annual Congress, Singapore 2027 | *ESMO Asia* Feb 2026

Associated with UT MD Anderson

- Project examines how pSTAT3, a cytokine-driven signaling pathway central to inflammation and tumor survival, reshapes the tumor microenvironment across six gastrointestinal cancers: esophageal, gastric, colorectal, pancreatic, hepatocellular, and cholangiocarcinoma.

Cholangiocarcinoma Foundation Conference 2027 — Utah | *Cholangiocarcinoma Foundation* Jan 2026

Associated with UT MD Anderson

Attendee, DAVANCO Conference | *DAVANCO* Jul 2025

Associated with UT MD Anderson

- Attended an invitation-only international oncology symposium in Bermuda; shadowed and followed an MD Anderson faculty delegate, one of whom is selected each year.

EDUCATION

Startup School 2026 | *Y Combinator* Jul 25–26, 2026

High School Diploma | *Cinco Ranch High School* 2024–2028

Katy, Texas

- Grade: Rising junior.
- Activities: Cancer Research Society president; Physics Club officer; Varsity Orchestra; NHS; SciOly invitational gold medal.

SELECTED PROJECTS

eBay Reselling Business | *Independent project*

- Scaled a business hobby to more than \$35,000 in net profit through eBay reselling.

Photography | *Independent business*

- Served 25+ clients; photographed proms, graduations, professional headshots, profile portraits, and landscapes. Portfolio: @sean.photodiary_.

VOLUNTEER & COMMUNITY LEADERSHIP

Summer Volunteer | Texas Children’s Hospital

Jun 2025–Jul 2025

Children

- Selected through a 4.3% acceptance process (13 of 300+ applicants) and served as a lead speaker.
- Completed 120 volunteer hours assisting nurses and staff by delivering supplies and transporting patients; supported patient care by assisting guests.

Volunteer | Cholangiocarcinoma Foundation

May 2026–Jun 2026

Social Services

- Set up conference sessions and special events and assembled 500 care kits at the Cholangiocarcinoma Foundation conference sent around the world.

Teaching Assistant | KCPCH Korean School Volunteering

Aug 2024–Present

Children

- Completed 150 volunteer hours, contributing approximately four hours per week.

Musician | Music for Medicine

May 2025–Present

Social Services

- Performs live music for elderly residents at care homes and assisted-living facilities.

Volunteer | Hope Plus Project / Hope to the Future Association

Jun 2025–Aug 2025

Social Services

- Designed SDG-themed shoes supporting children’s health rights in South Sudan.
- Profile notes certification by an Undersecretary of the Ministry of Youth and Sports, Republic of South Sudan.

SKILLS

Visible LinkedIn skill labels

Survival Modeling

Statistical Modeling

Research Writing

TCGA Data Analysis

Shadowing

Marketing

Adobe Lightroom

Research competencies

Computational cancer genomics

Translational oncology

Multi-omic analysis

Spatial transcriptomics

Transcriptomics

Genomics

Tumor-microenvironment analysis

Systematic review

Meta-analysis

Clinical and molecular data analysis

Radiomics

Scientific presentation

Research mentorship

ADDITIONAL PROFILE DETAILS

- Based in Katy, Texas; LinkedIn displays 500+ connections and 1,124 followers at the time of capture.
- LinkedIn lists 27 total skills; only the skill names visible in the supplied screenshots are reproduced above.
- Featured profile items include a Journal of Clinical Oncology supplement link, “TCH Junior Volunteers 2026” on YouTube, and a Y Combinator Startup School 2026 post.